Project Report

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Subject Name: IOT LAB
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Aim: IoT Based Smart Home System to control and monitor home lights from a remote location using Node MCU .

Objectives:

- To control and monitor household appliances from a remote location using the Internet.
- To switch appliances between manual control and automation.
- To integrate voice control using Google Assistant or Amazon Alexa.
- To control multiple appliances using a relay module and NodeMCU.

Components Required:

S.No	Name of Component	Quantity
1	NodeMCU (ESP8266)	1
2	LED (Bulb representation)	1 or 2
3	Breadboard	1
4	Resistor (220Ω – 330Ω)	2
5	Micro USB Cable	1
6	Jumper Wires	As required
7	Smartphone (for control)	1

Details of Components:

NodeMCU:

Microcontroller with built-in WiFi
Operates at 3.3V
Used to control devices via internet

LED :

Acts as a bulb/device
Glows when powered
Controlled through GPIO pins

Resistor (220Ω–330Ω):
Limits current to protect LED
Prevents burning of LED

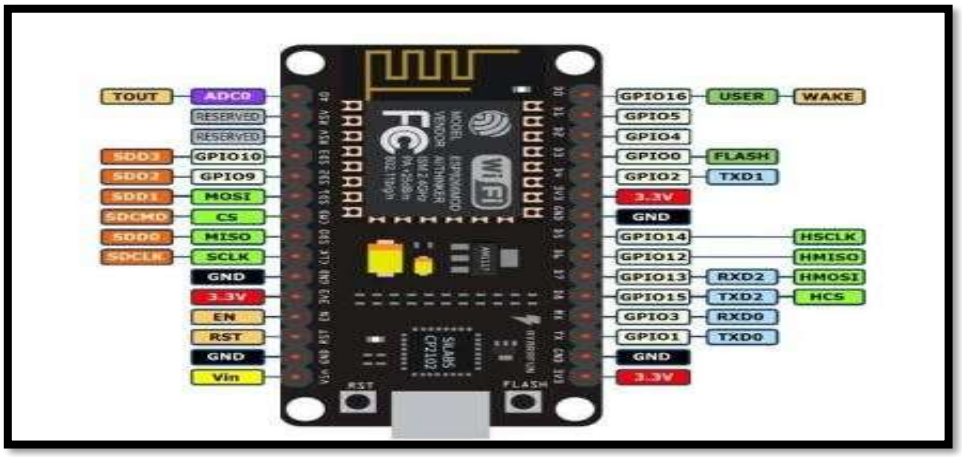


Figure 1 : NodeMCU Board

4. readboard & Jumper Wires

Used for temporary circuit connections

No soldering required

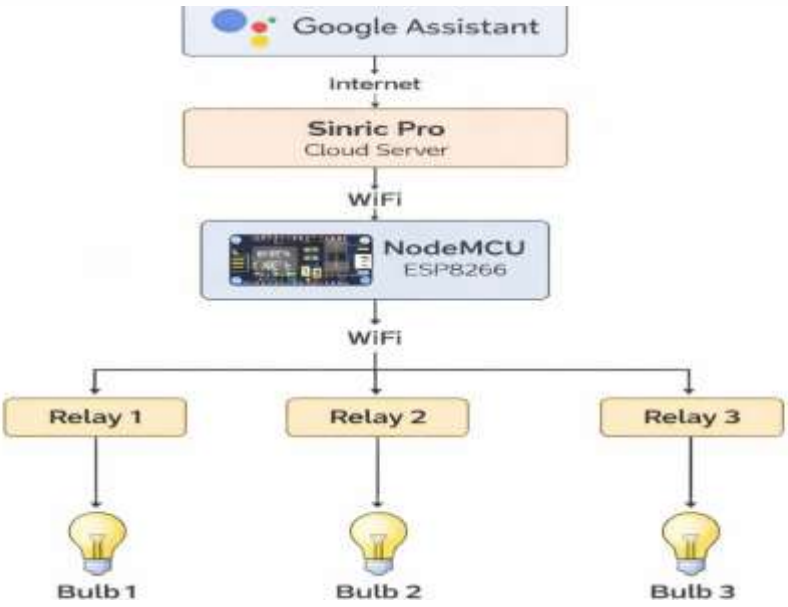
Block Diagram of Designed Model:

The block diagram illustrates the interconnection between system components.

Main components include:

- NodeMCU (ESP8266) – main controller
- Sinric Pro Cloud Platform – used for remote and voice control

The NodeMCU receives commands from the cloud platform and activates the relay channels to control the connected bulbs.



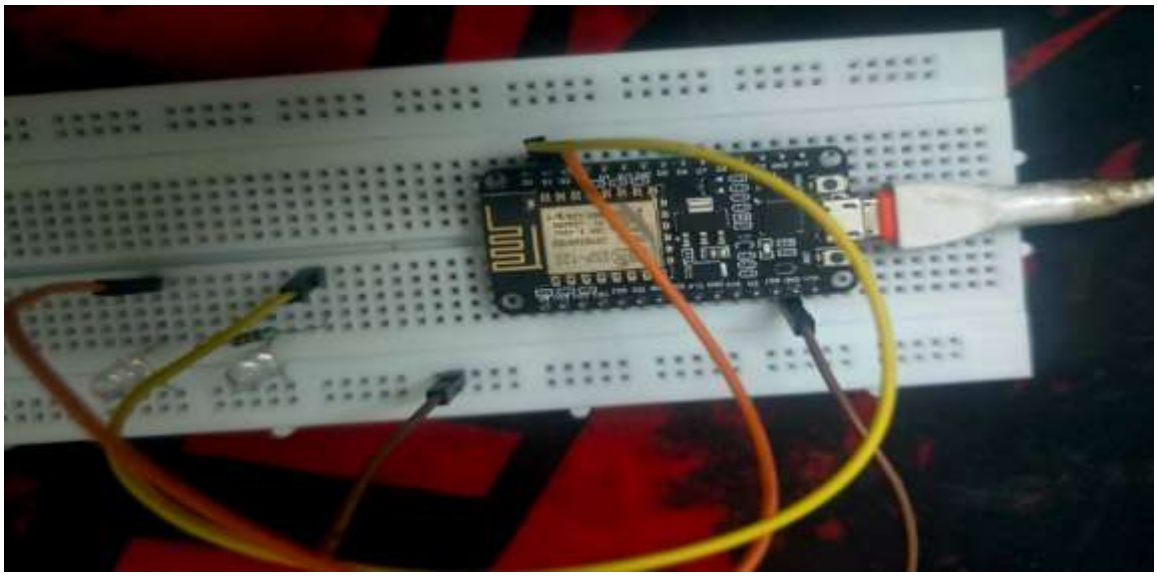
Working of Designed Model:

The **NodeMCU (ESP8266)** is connected to a WiFi network using SSID and password.
The device is linked to the **Sinric Pro platform**, which acts as a cloud server.
The user sends commands (ON/OFF) from a smartphone using:

- Google Assistant / Alexa / Sinric Pro App

The command goes to **Sinric Pro cloud**, then to the NodeMCU via internet.
The NodeMCU processes the command and controls the GPIO pin (like D1 or D2).
The LED connected to that pin turns ON or OFF accordingly.

8. Pictures of Prototype



The above image shows the physical prototype of the smart home automation system. It includes:

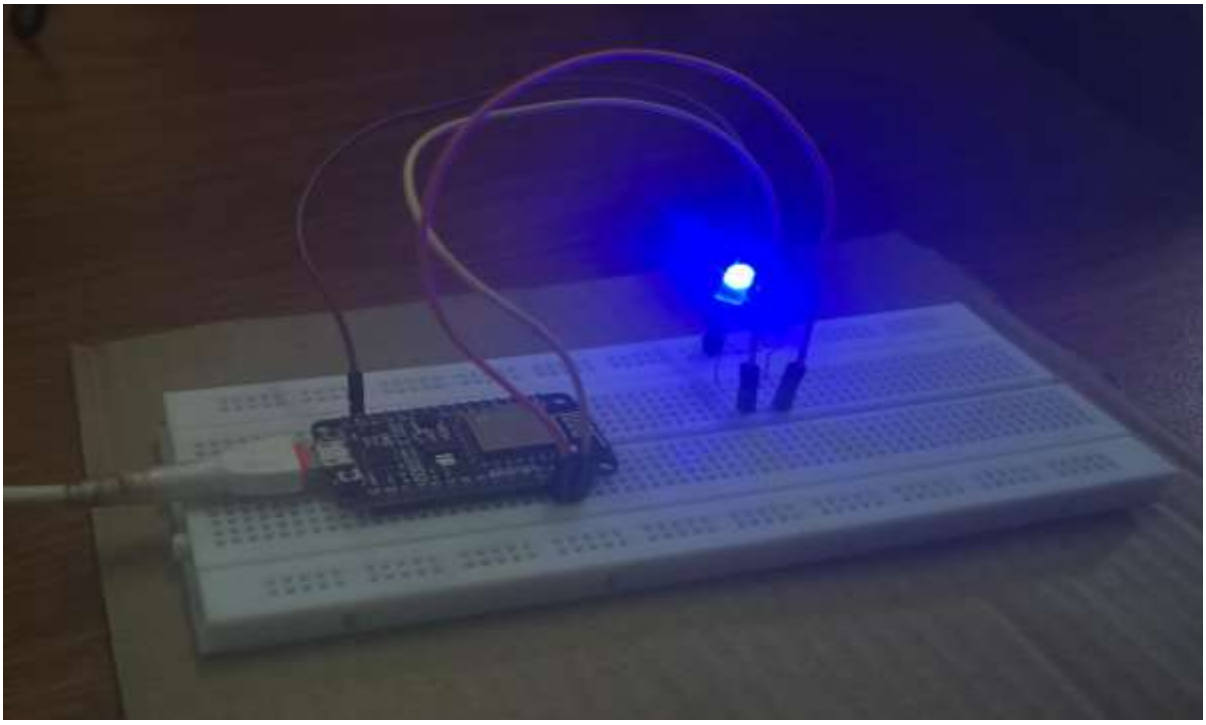
- NodeMCU microcontroller
- Two bulbs connected through a relay module

This demonstrates the working IoT-based smart home model.

9. Output of Designed Model / Prototype

This section typically includes:

- Screenshots from the mobile app interface (Sinric Pro)
- Voice assistant control demonstration



Learning outcomes :

- Gained practical experience in developing IoT-based systems using NodeMCU (ESP8266).
- Learned how to integrate cloud platforms like Sinric Pro for remote device control.
- Understood the concept of voice-controlled automation using Google Assistant or Alexa.
- Developed skills in interfacing hardware components such as relay modules, and with microcontrollers.
- Learned how to control multiple appliances using a multi-channel relay module.
- Gained knowledge of basic energy monitoring and electricity bill estimation in IoT systems.
- Improved practical skills in testing, troubleshooting, and implementing smart home automation systems.